In-Hand Object Classification and Pose Estimation With Sim-to-Real Tactile Transfer for Robotic Manipulation

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***Abstract*—Dexterous robotic grasping gains great benefits from tactile sensation, but delicate object exploration by tactile information is challenged by difficulty in rich and efficient data production. In this letter, we propose a tactile-based in-hand object perception approach, empowered by a sim-to-real approach toward a data-efficient learning process. A high-fidelity tactile input, measured by a pair of vision-based tactile sensors, was represented as a point cloud facilitating dual functionality of object classification and the associated pose estimation. For the classification, we constructed *PoinTacNet*, a variation of *PointNet* to fit into tactile data processing. A reliable simulation methodology on tactile input was employed for the pretraining of the model, transferred to the fine-tuning process facing real tactile data. Taking inspiration from human behaviors, a re-grasping strategy was imparted by means of conditional accumulation of class likelihood distribution. The result of the framework facilitates a high object classification accuracy of 83.89% on the ten objects from McMaster-Carr's CAD models, which is significantly improved by the re-grasping. In addition, a set of benchmarks displays the computational efficiency in the sim-to-real transfer. In line with the successful classification, the posture of in-hand objects is estimated using point cloud registration algorithms, achieving an average angular and translational RMSE of 5.03° and 2.41 mm, respectively. The proposed approach has the potential to enable robots to attain human-like haptic exploration skills for perceiving unstructured environments.**

***Index Terms*—Perception for grasping and manipulation, deep learning in grasping and manipulation, transfer learning.**

I. Introduction

THE human hand plays a substantial role in delicate physical interactions with objects. Manipulating objects with hands incorporates haptic exploration to identify their type, shape, and configuration, whether they are known or unknown, by leveraging the collaborative and synergistic sensations of both visual and tactile feedback [1], [2]. Among these, the distributed tactile sensation provided by dense mechanoreceptors is essential for characterizing the mechanical properties of the objects, which cannot be accomplished through visual feedback alone. Humans are able to perceive and understand the objects they manipulate through tactile information collected from their hands in a variety of specific exploratory procedures [3], [4]. Based on this afferent information, people continuously learn the governing behavioral strategies throughout their lives to generalize and improve the performance of tactile exploration.

For robots to handle objects with dexterity, a reliable and accurate understanding of the objects is essential. Recently, there has been a growing interest in the field of robotics to imitate human haptic exploration, aiming to improve the performance and versatility of robotic manipulation systems. Alongside the emergence of various computer vision techniques, exploration procedures on robotic systems have typically been accompanied by vision or laser sensors [5], [6], [7]. However, relying solely on vision sensors for object identification exhibits drawbacks, such as a loss of reliability of vision due to low-light conditions or obscuring of in-hand objects by the surroundings of a robot manipulation system [8]. On the other hand, in unstructured environments where the mechanical information of objects is essential but unpredictable through visual cues, robots require haptic exploration involving direct interaction with the objects. These challenges have been overcome through the development of tactile sensors and advancements in artificial intelligence techniques [9], [10], [11], [12]. While previous research has made significant progress in robotic exploration, a common limitation of using deep learning methods is the requirement for extensive prior exploration with tactile sensors, which can be costly and time-consuming.

In this letter, we proposed a framework that can classify and estimate the pose of in-hand objects using vision-based tactile sensors, GelSlim [13], as illustrated in Fig. 1. This framework drew inspiration from the powerful strategy employed by humans to understand in-hand objects without visual feedback. Our proposed method employed the tactile sensor to extract tactile information as a point cloud that effectively represents the 3D shape of an object's local region. We also developed a tactile perception model based on *PointNet* [14] called *PoinTacNet* that can classify in-hand objects based on tactile information. Our tactile perception model was initially trained with simulated tactile data. Subsequently, real tactile data was integrated using transfer learning with the pre-trained model, enabling the model

Manuscript received 28 June 2023; accepted 6 November 2023. Date of publication 20 November 2023; date of current version 1 December 2023. This letter was recommended for publication by Associate Editor N. Figueroa and Editor J. Kober upon evaluation of the reviewers' comments. This work was supported by Electronics and Telecommunications Research Institute (ETRI) funded by the Korean Government, through A Study of Hyper-Connected Thinking Internet Technology by autonomous connecting, controlling, and evolving ways under Grant 23ZR1100. *(Corresponding author: Jung Kim.)*

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This letter has supplementary downloadable material available at https://doi.org/10.1109/LRA.2023.3334971, provided by the authors.

Digital Object Identifier 10.1109/LRA.2023.3334971

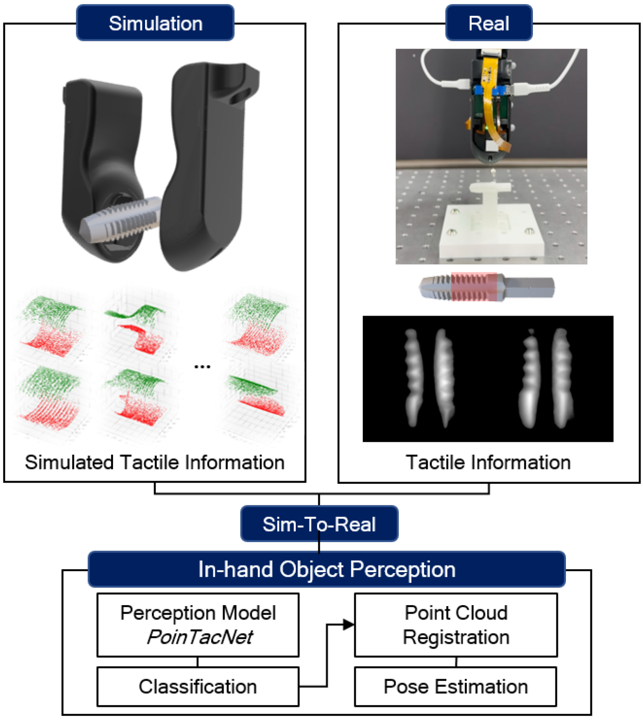

Fig. 1. Our proposed framework for perceiving in-hand objects through tactile information. Tactile information extracted from real and simulation was used to train the perception network called *PoinTacNet*. Object classification tasks can be enhanced with the implementation of humans' haptic exploration strategy (re-grasping method). Using the point cloud registration algorithm, the pose of the in-hand object is estimated.

to adapt its understanding of tactile sensations from simulated environments to real-world scenarios. We demonstrated an improvement in the performance of in-hand object perception by emulating the humans' exploration strategy called re-grasping (details in Section III-E). In addition, the FilterReg [15] algorithm was used to perform the pose estimation tasks. The contributions of this letter are summarized as follows:

1) Provides an approach for accurate classification and poses estimation of in-hand objects without reliance on visual information.
2) Reduces the domain gap between simulation and real-world tactile information by applying transfer learning on the proposed network.
3) Proposes the re-grasping strategy that enhances certainty in identifying in-hand objects, which involves multiple touches to gain a more definitive understanding of their identity.

II. Related Works

A. In-Hand Objects Perception Using Real Tactile Data

The research on understanding in-hand objects with artificial intelligence techniques and tactile sensors typically involves data collection through exploration procedures in real-world settings [16], [17], [18], [19], [20], [21], [22]. For instance, Luo et al. [16] proposed an approach called Iterative Closest Labeled Point (iCLAP) that can integrate tactile and kinesthetic information for object recognition by recording the 3D location of the tactile sensor and tactile reading simultaneously. Spiers et al. [17] proposed a method for acquiring useful tactile features and identifying objects with a single grasp utilizing a two-finger adaptive robotic gripper equipped with array-type tactile sensors known as *TakkTile* [23]. Despite the limitations of time, computation, and hardware, the authors emphasized that the ability to acquire useful tactile information during simple and functional actions can extend the applicability of tactile sensing. Using parametric estimation methods and a machine learning technique, the authors were able to classify objects and extract their dimensions, stiffness, and pose within error bounds. Pastor et al. [19] distinguished grasped objects using a squeeze-and-release motion and a high-resolution tactile sensor that was integrated into a gripper. Sequences of pressure images along the applied gripping force formed a 3D tensor that was fed to a classifying 3D CNN called *3D TacNet*. Bauza et al. [20] proposed a tactile pose estimation method using the first haptic glance for known objects. This was achieved by learning the perception model directly from the object geometry, obtained through a set of tactile imprints collected with the vision-based tactile sensor GelSlim. Combining tactile imprints with robot kinematics, a tactile model of an object for localization was created. Roberge et al. [22] proposed the use of tactile sensing to identify objects since tactile signals can detect physical properties, such as stiffness and texture, that vision sensors may miss. The authors proposed a rapid, grasp-oriented tactile exploration procedure for the identification of a large datasets of commonplace objects. Through the three phases of tactile exploration (contact, rubbing, and compression), capacitive tactile sensors recorded three modal tactile signals to classify the objects. The aforementioned research needs extensive prior exploration and tactile feature extraction processes with tactile sensors, which is expensive and time-consuming.

B. In-Hand Objects Perception Using Simulated Tactile Data

To address the problems of expensive costs of real-world data collection, other works, more reminiscent of our approach, have acquired tactile experience through simulation [24], [25], [26]. Dikhale et al. [24] presented a network architecture based on pixel-wise dense fusion to combine vision and tactile data for the estimation 6D pose of in-hand objects. Since the estimation of in-hand objects' pose includes occlusions due to the robot's gripper or other objects, the author emphasized the use of tactile data that can improve its estimation and therefore fuses the tactile data in the point cloud form with vision data. The authors presented the synthetic data collection procedures through simulation. The authors demonstrated the outperforming estimation results under heavy occlusion with tactile and vision data fusion. Villalonga et al. [25] conducted a follow-up study on prior research [20] that did not necessitate prior exploration. The authors proposed a framework for estimating the pose of an in-hand object using a perception model learned with simulated tactile information based on geometric contact rendering and contact shape matching using contrastive learning tools. This work localized the object from the first touch and reasoned over pose distributions rather than providing a single pose estimate result.

Our framework allows robots to perceive objects without extensive prior tactile experience using a sim-to-real approach. Additionally, unlike several previous studies that focus only on the in-hand object pose estimation task, our approach allows for classification and pose estimation of in-hand objects in a sequential manner.

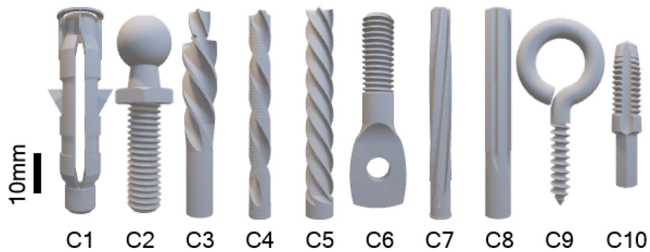

Fig. 2. Ten objects selected, which are available on McMaster-Carr [27]. From the left, C1 (Anchors), C2 (Ball stud), C3 (Counterbore), C4 (Drill bit), C5 (End mill), C6 (Machine hangers), C7 (Hand reamer), C8 (Shank reamer), C9 (Routing eyebolt), and C10 (Shank tap).

III. Methods

A. Objects Selection

The proposed framework is applied to ten objects selected from CAD models, which are available on McMaster-Carr [27] (Fig. 2). We intentionally included objects with similar shapes (e.g., C4, C5, C7, and C8) that are difficult for vision sensors to distinguish. Each object has unique shape characteristics, such as screw lines and protrusions, and an aspect ratio ranging from 2.01 to 9.52.

B. Tactile Data Generation Via Simulation

This work adopts a simulation-based approach for generating tactile datasets, leveraging the MuJoCo simulator [28]. Tactile information is simulated in the form of depth maps, rather than high-resolution RGB images. Using depth images directly provides spatial information about the object, including a clear distinction between grasped and non-grasped regions, while also avoiding unnecessary complexity introduced by simulating RGB images. Given the 3D model of objects aligned along the z-axis, two depth cameras are placed to face the object in opposite directions with respect to the z-axis, as illustrated in Fig. 3(a). Subsequently, artificial prismatic and cylindrical joints are added on two depth cameras to fully scan the outer of the object. The viewpoint transformation by each joint is governed by rotation to angle α, and translation to displacement β, respectively. To render the resultant depth image, the threshold that can consider only pixels between distance d and Δd is set. The overall operation was performed in Python (using MuJoCo-py library.) The value of variables was set as follows:

$$\begin{aligned} &\alpha \in \{i \mid 0 \leq i \leq 360, i \in \mathbb{Z}\} \\ &\beta \in \{0.005j \mid -9 \leq j \leq 10, j \in \mathbb{Z}\} \\ &d = 6\,\text{mm}, \qquad \Delta d = 2\ \text{mm} \end{aligned} \tag{1}$$

MuJoCo uses OpenGL that non-linearly normalizes real depth value into a value called *z-buffer* within the range of [0, 1]. Therefore, the *z-buffer* needs to be transformed using (2), and (3) relating between the real depth (z_e) and the value in the *z-buffer* (z_b) in order to obtain the real depth value from the produced depth image in MuJoCo. First, it has to be converted into normalized device coordinates (NDC, ranging in $[-1, 1]$, z_n) which is a normalized value between the nearest clipping plane (z_{near}) of -1 and the farthest clipping plane (z_{far}) of 1 by (2) [29].

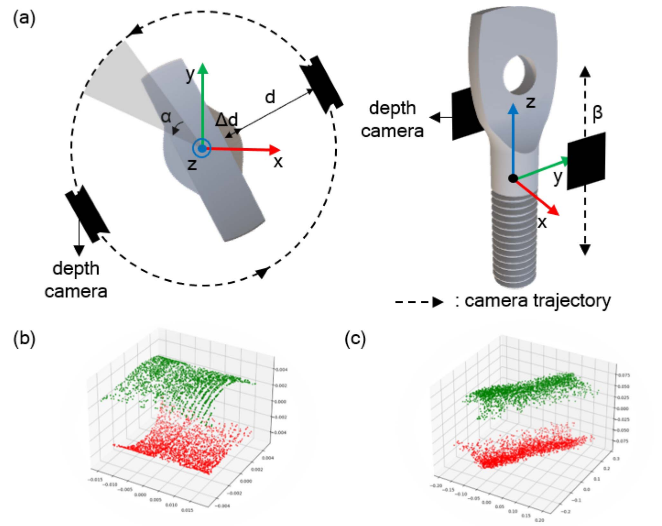

Fig. 3. (a) Depth cameras configuration in MuJoCo simulator. 3D model of the object is fixed on origin of the MuJoCo simulator. (b) Raw tactile point cloud data and (c) noised, randomized, rotated, and normalized tactile point cloud data.

$$z_n = 2z_b - 1 \tag{2}$$

Equation (3) can be used to retrieve the real depth value z_e after transforming z_b into z_n, undoing the non-linear mapping.

$$z_e = \frac{2z_{near}z_{far}}{z_{far} + z_{near} - z_n(z_{far} - z_{near})} \tag{3}$$

After rendering depth images from each object in real depth value z_e, the depth images are converted into point cloud through Open3D [30], an open-source Python library. Open3D converts all pixels in a given depth image into a point cloud. We randomly selected 2048 points from the converted point cloud of the depth image for computational efficiency. Fig. 3(b) shows an example of rendered tactile point cloud data.

C. Tactile Perception Model, PoinTacNet

Fig. 4 illustrates the architecture of the proposed neural network, *PoinTacNet*. *PoinTacNet* is employed for classifying in-hand objects based on tactile information extracted from two GelSlims, with each input having a shape of (2048, 3). In designing the network architecture, we incorporated an intuition drawn from human haptic exploration: the more partial features provided, the more confidently humans can identify the object. The network consists of three phases: the 'Individual Touch (IT)' phase, the 'Certainty Amplification (CA)' phase, and the 'Touch Analyzer (TA)' phase. The IT phase encodes the raw tactile signal. Two inputs are utilized in this phase for the use of parallel grippers in real experiments. The 'Input transform' in this phase refers to a learned affine transformation matrix, a concept originally introduced in *PointNet*. As the network gets deeper in this phase, the number of channels are increased to facilitate the extraction of high-level features from the raw tactile data (refer to Fig. 4; in the IT phase, the channel counts are 3, 16, 32, 64, and 256). This enables the network to encompass a broader spectrum of detailed information pertaining to the

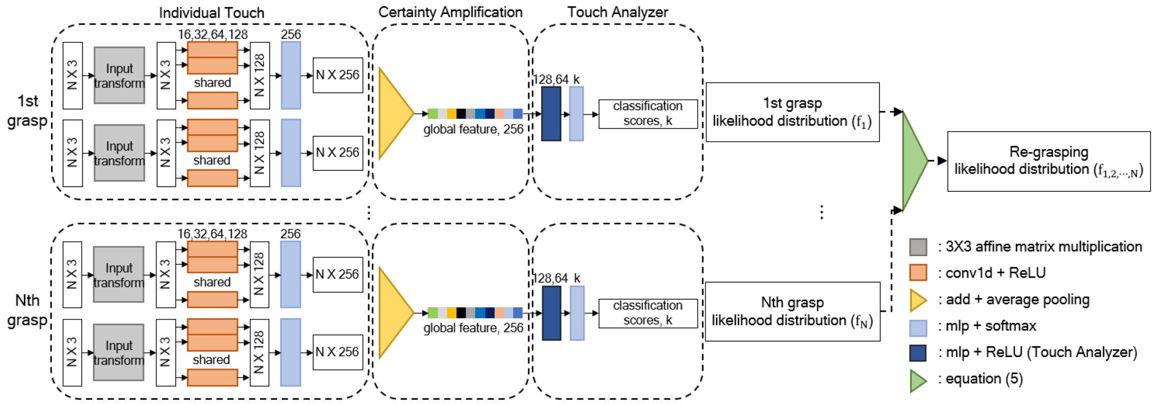

Fig. 4. Architecture of proposed perception network called *PoinTacNet* and process of implementing the re-grasping method. *PoinTacNet* consists of three phases (IT, CA, and TA). Object class likelihood distribution is accumulated through (5) that can enhance the perception of in-hand objects.

explored object. The CA phase serves as a crucial intermediary step between the primary phases. In this phase, both add and average-pooling operations are applied to the class likelihood obtained from the Softmax function at the end of each touch in the IT phase. This dual pooling approach is intentional; it aims to enhance the probability distribution of unique features associated with the object. Specifically, it leverages the probabilities derived from individual sensors' features through the Softmax in the IT phase, and by combining them in the CA phase, it effectively augments the certainty of the object's distinctive features. This design choice seeks to reinforce the representation of key object attributes for more accurate classification results. The third phase is called TA, which analyzes the tactile features extracted from the IT phase. In this phase, the number of channels decreases as the network deepens (see Fig. 4; in TA, the numbers of channels are 256, 128, 64, and k, where k is equal to the number of classes). By reducing the number of channels, the network simplifies the representation of features. This helps in concentrating on the most relevant information for the classification task, which is essential for accurate object identification. A 256-size of global feature is fed into the TA phase. In more detail, this letter uses the ReLU activation function in every phase and dropout layer with a rate of 0.3 in the TA phase.

For the training of *PoinTacNet*, we utilized the public TensorFlow 2.4.0 and ran it on an Intel(R) Xeon(R) Gold 6130 and TITAN Xp GPU. Adam was used as the optimizer, with a learning rate of 0.0005 and a batch size of 256. In order to prevent over-fitting on the training datasets, we employed early stopping as regularization. As a role of domain randomization, Gaussian noise was imparted to the input, normalized with the gripper's maximum opening length, and the tactile data was randomly rotated with respect to the z-axis of the training data to ensure robust classification at any gripper angle (refer to Fig. 3(c)). The total amount of point cloud data available for training the networks was 687 k, with 85 k data points allocated for both the validation and test sets.

D. Transfer Learning Using Real Tactile Data

Since GelSlim estimates the depth map by backtracking the deformation of elastomer and light intensity [31], there exist smooth edges that generate the gradients around the in-contact indentation, unlike depth cameras in simulation. This gap between the real world and the simulation can be reduced by introducing the real tactile knowledge to *PoinTacNet*, which is pre-trained completely through simulated tactile knowledge. A reduced learning rate is used to fine-tune the weights based on a model that has been pre-trained to learn from real tactile data. A Kinova Gen2 robotic arm equipped with a Robotiq F-85 gripper and two GelSlims attached to the fingertips is controlled by a Robot Operating System (ROS) Python script to capture the real tactile point cloud datasets. The configuration for collecting real tactile point cloud data is illustrated in Fig. 5. For stable and consistent data collection, a small pillar was added to each object in their 3D CAD model to keep them stationary between grasps. The target position of the Kinova manipulator's end-effector relative to the object's reference frame is expressed as an (4).

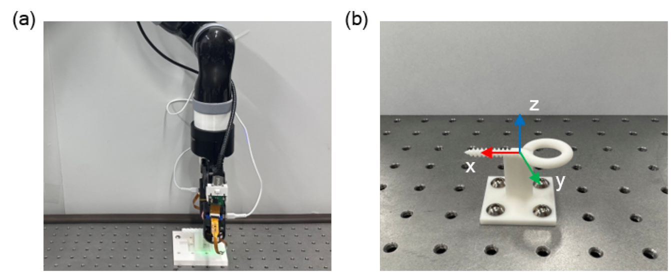

Fig. 5. (a) Experimental setup to collect real tactile point cloud for applying transfer learning on *PoinTacNet*. (b) 3D-printed object is fixed to the base of environments.

$$x \in \{0.001r \mid -30 \leq r \leq 30, r \in \mathbb{Z}\},$$
$$z, y = 0 \tag{4}$$

Each object was grasped six times at each end-effector position. Four grasps' tactile point cloud data were utilized for transfer learning, while the others were allocated for validation and test data.

During the fine-tuning process, the IT phase was frozen and not trained. The amount of real tactile data used for the train was 1.8 k in total, while 0.4 k were used for validation and test each, with the same batch size of 256. By utilizing a small amount of real tactile point cloud data, the weights of the CA and IT phases were tuned with a reduced learning rate value of 0.0001

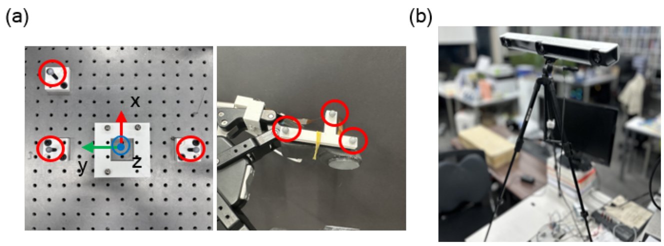

Fig. 6. Experimental setup for the pose estimation tasks. (a) Six markers were attached to the base of the robot environment and gripper to measure reference and gripper frames. (b) V120 : TRIO was used to measure the ground truth of V values.

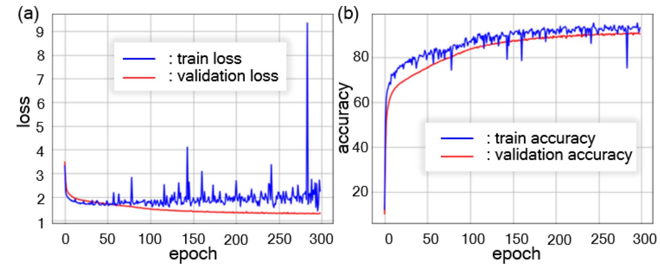

Fig. 7. Learning curve of *PoinTacNet* during training epochs. (a) and (b) show the average loss and accuracy curves during epochs, respectively.

compared to the learning rate that was used with simulated tactile point cloud data.

E. Enhancement of In-Hand Object Perception, Re-Grasping

The re-grasping method embodies humans' effective tactile exploration strategies, which involve touching and feeling an object multiple times to gain more certainty about its identity and position. By leveraging the multi-contact knowledge from [20], the re-grasping method utilizes the likelihood accumulation strategy, as (5)

$$\begin{aligned} f_{1\ldots N} &= P(y|G_1,\ldots,G_N) \\ &= \frac{\Pi_{i=1}^{N} P(y|G_i)}{\sum_{y'=1}^{N}\Pi_{i=1}^{N} P(y'|G_i)} \\ &= \frac{f_1 \otimes f_2 \otimes \cdots \otimes f_N}{\mathrm{sum}(f_1 \otimes f_2 \otimes \cdots \otimes f_N) + \epsilon} \end{aligned} \tag{5}$$

where G_i and y are tactile input at ith grasping and object class respectively, and $P(y|G_i)$ represents the likelihood distribution over object class at ith grasping, abbreivated to f_i for simplicity. $\epsilon(= 1\mathrm{e}^{-6})$ is used for preventing zero-division. $\otimes$ represents element-wise multiplication.

F. Pose Estimation

With the classified point cloud set extracted from tactile sensors, the grasped object's pose can be estimated through the point cloud registration algorithm. In this letter, we used a probabilistic point cloud registration algorithm, FilterReg. FilterReg proposes a probabilistic method that transforms the registration problem into a maximum likelihood estimation and solves it using the EM algorithm, which is faster and more reliable to noise, outliers, and occlusions. The source and target point clouds are considered as local point clouds obtained from the contact point between the tactile sensor and 3D modeling of the in-hand object, respectively.

The objective of pose estimation in this work is to identify the object's 2D position (t_x, t_y) and 1D angle (θ) with respect to the surface of the sensor's gel rather than the full 6D pose, given the assumption that the grasping method of the parallel gripper always involves grabbing the side of the chosen objects. Fig. 6 shows the setup used for the pose estimation tasks. The Kinova robot arm and Robotiq gripper were controlled using ROS Python scripts. The gripper was rotated in increments of $-15°$, $-10°$, $0°$, $10°$, and $15°$ with respect to the y-axis of the reference frame, which represents the stable rotation range of the gripper. However, due to inaccurate rotation of the gripper compared to the desired values, V120 : TRIO from NaturalPoint, Inc. DBA OptiTrack was utilized to measure the actual poses of the gripper. Three markers were attached near the base of the environment to define the reference frame, while three additional markers were attached to the gripper, and V120: TRIO allows the measurement of the positions of the reference and gripper markers. The processing of the gripper and reference frames was carried out using MATLAB to measure the ground truth of poses.

IV. Experiments and Results

A. Tactile Data Classification Results

The learning curve for *PoinTacNet* in Fig. 7 shows that it performs exceptionally well in classification tasks on both the simulated train and validation datasets. The average classification accuracy for the train and validation datasets was 91.1% and 95.4%, respectively.

Real tactile point cloud data is transferred to the pre-trained *PoinTacNet*. The results show that *PoinTacNet* is able to classify real tactile point cloud data despite the domain difference between the real-world tactile sensors and the depth cameras in MuJoCo. The normalized confusion matrix is shown in Fig. 8(a). The results indicate that *PoinTacNet* exhibited strong performance for every class, achieving an average accuracy of 83.89%. The lowest ranked class, C2, achieved a correct labeling rate of 61.3%, and the highest misclassification errors occurred when misclassifying C2 as C8, with an error rate of 19.4%, followed by misclassifying C4 as C7, with an error rate of 17.3%. Secondly, *PoinTacNet* was trained solely with the real tactile point cloud datasets, and the confusion matrix is presented in Fig. 8(b). The overall classification accuracy achieved was 49.01%, with C1 having the highest accuracy of 93.90%, while C2 and C10 had a accuracy of 0%.

The high-dimensional embeddings were visualized in a three-dimensional space using a dimensionality reduction technique, t-SNE. This was done with simulated data, real tactile data using transfer learning, and real tactile data without transfer learning. The simulated and real tactile data were clustered well after applying for sim-to-real tactile transfer, as shown in Fig. 9(a) and (b). However, without tactile transfer, the embeddings were not effectively clustered in the feature space, as shown in Fig. 9(c).

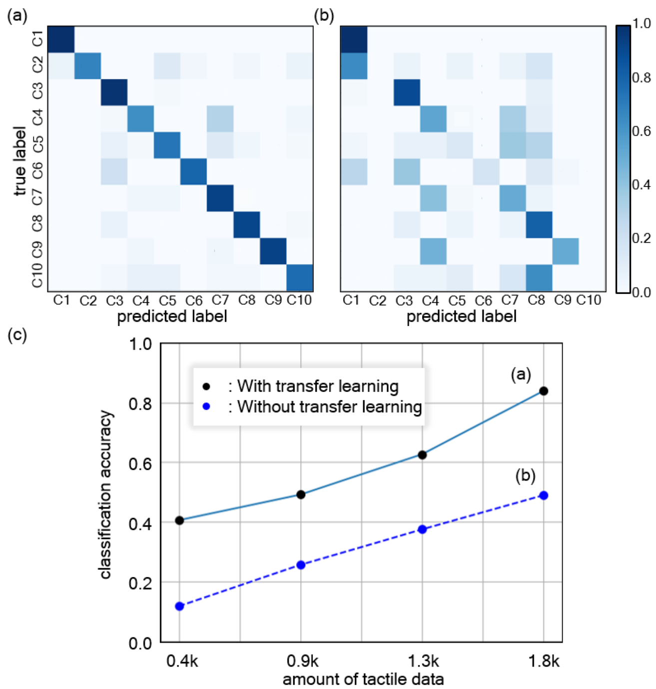

Fig. 8. (a) Obtained normalized confusion matrix of *PoinTacNet* tested on the real tactile point cloud datasets after applying transfer learning. (b) The obtained normalized confusion matrix of *PoinTacNet* tested on the real tactile point cloud datasets after trained with only the real tactile datasets. (c) The overall classification accuracy of *PoinTacNet* trained with real tactile point cloud datasets.

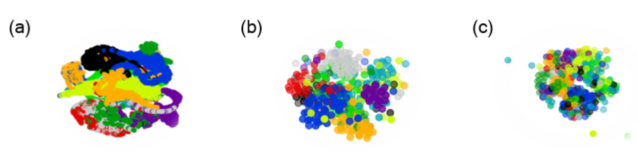

Fig. 9. Comparison of t-SNE visualization with (a) simulated tactile data, (b) real tactile data with transfer learning, and (c) real tactile data without transfer learning.

B. Evaluation of Transfer Learning Efficiency

In this section, an experiment was conducted to determine how much data is required to train *PoinTacNet* solely with the real tactile datasets without transfer learning. The experimental result is depicted in Fig. 8(c). *PoinTacNet* could achieve the classification accuracy of 83.89% even with a small amount of real tactile data of 1.8 k. However, with the same amount of data, it only exhibited a classification performance of 49.01%, when trained with real tactile data only. The results show that the proposed sim-to-real transfer method can largely enhance data collection efficiency, at the same time ensuring effective object classification.

C. Re-Grasping Results

The powerful haptic exploration strategy of humans is implemented via the re-grasping method, as illustrated in Fig. 4. Humans accumulate class likelihood through re-grasping techniques. The class likelihood in each grasp is derived from this intuition using (5). Fig. 10 demonstrates that overall accuracy

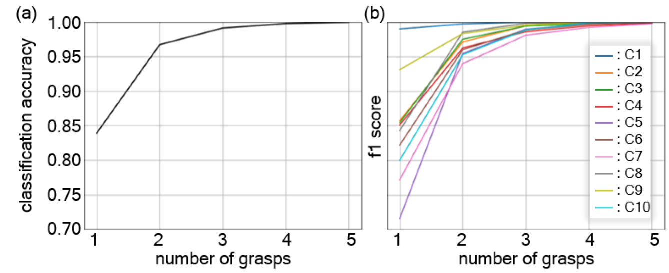

Fig. 10. (a) Overall accuracy changes over the number of grasps. (b) F1 scores change on each object over the number of grasps.

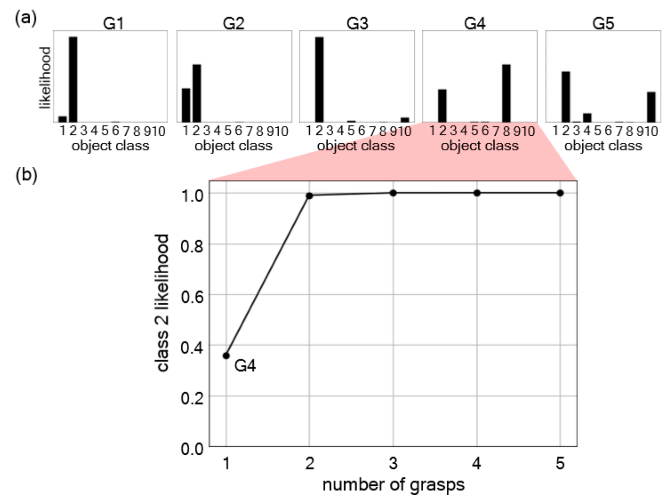

Fig. 11. (a) Class likelihood distribution on every single grasp. (b) Object (class 2) likelihood changes as the number of re-grasping times increases.

and F1-score increase as the number of re-grasping times increases across all classes, where F1-score is a classification performance metric and a weighted average of precision and recall. Particularly, beginning with the second re-grasping, the overall accuracy and F1-score were improved dramatically.

Also, the experiment demonstrates that misclassification of grasp can be corrected by re-grasping. Each grasp's class likelihood is illustrated in Fig. 11(a), and grasp 1,2,3,5 (G1, G2, G3, G5) correctly classified C2, with the exception of grasp 4 (G4 in Fig. 11(a)). Grasp 4 incorrectly categorized C2 as C8. In the case where G4 is the initial grasp, the misclassification was rectified through re-grasping. Fig. 11(b) illustrates the results of re-grasping the incorrectly classified Grasp 4. As the number of re-grasps increased, the class likelihood distributions indicate that the object was correctly classified. From the beginning of the second grasp, the certainty of the object in hand increased significantly. In the first grasp, the certainty of C2 was approximately 0.357, but in the second grasp, it increased to approximately 0.990. This indicates that with re-grasping, the certainty of the object in hand was emphasized.

D. Pose Estimation Results

As described in Section III-F, the point cloud registration algorithm, FilterReg, was used to estimate the pose of objects held in hand. As shown in Table I, the RMSE value of each class's estimated pose value was determined. The robot achieved

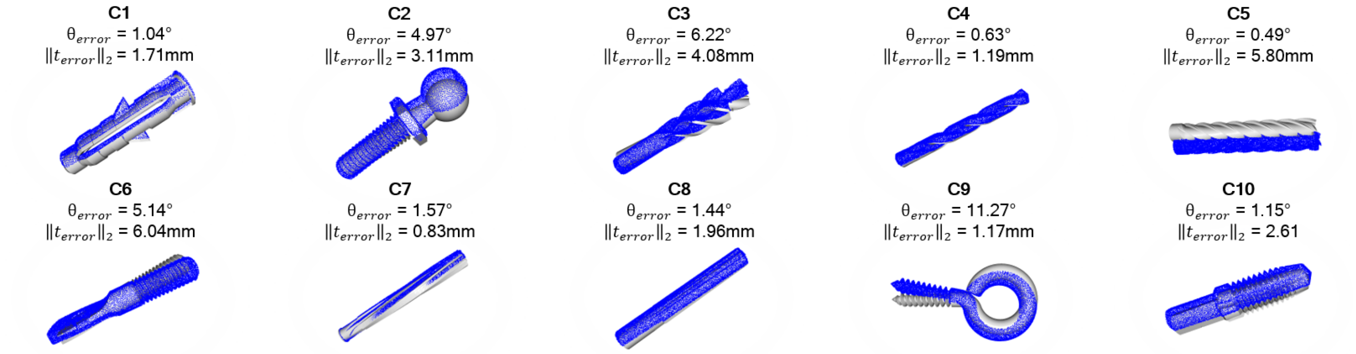

Fig. 12. Examples of pose estimation results on each object. θ_{error} represents the error between the measured θ obtained through V120 : TRIO for each object and the estimated θ derived via the point cloud registration algorithm. $\|t_{error}\|_2$ denotes the norm of the translational error in the x and y directions.

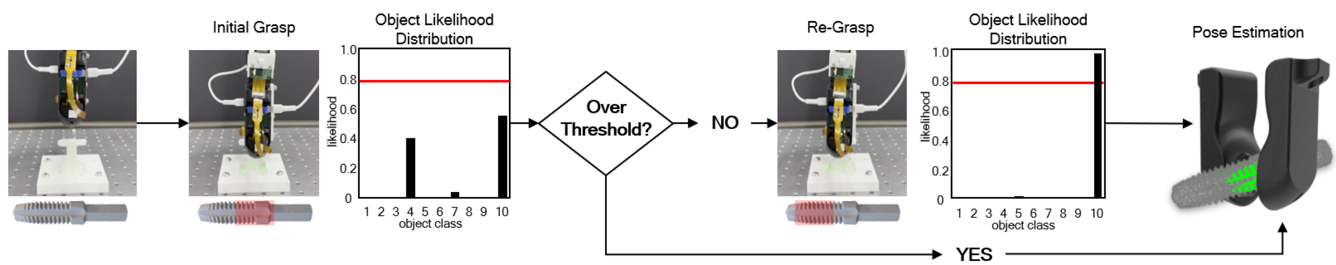

Fig. 13. Scheme of demonstration in re-grasping strategy for robotic exploration procedures.

TABLE I
RMSE Value of Pose Estimation Results on Each Class

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
θ (°)	1.18	10.63	8.13	1.95	3.00	6.21	2.77	2.89	11.72	1.84
$t_x(mm)$	0.50	2.15	3.31	0.40	0.31	3.44	0.08	1.04	0.38	1.84
$t_y(mm)$	1.48	0.88	0.58	0.48	1.96	1.68	0.50	0.45	0.78	0.86

an average RMSE of 5.03° for angular errors and an average translation error of 2.41 mm. For the angular error, C9 had the highest RMSE value of 11.72°, while C1 and C10 demonstrated the lowest values of 1.18° and 1.80°, respectively. For the translational error, t_x was consistently larger compared to t_y, and t_x exhibited a similar trend to the results of θ. Since the portion of the local point cloud and global point cloud affects the results of point cloud registration, small objects with an elongated shape tend to show low errors typically. However, objects that have relatively large protrusions than elongated parts, such as C2, C6, and C9, show high errors. Examples of pose estimation results on each object are shown in Fig. 12.

E. Demonstration

We demonstrated a robotic exploration leveraging the knowledge of the re-grasping strategy as shown in Fig. 13. The initial grasp generates a distribution of class likelihood that can determine the certainty of the in-hand object based on a manually set threshold (set to 0.8, the red line in Fig. 13). If the certainty of the in-hand object does not exceed the threshold, the gripper re-grasps another part of the object to collect the class likelihood distribution until the certainty exceeds the threshold. Once the threshold is exceeded, the posture of the object is estimated using the point cloud registration algorithm. The demonstration illustrates how the knowledge of an effective human haptic exploration strategy by re-grasping enhances the perception of in-hand objects.

V. Conclusion

This letter proposed a classification and pose estimation method for in-hand objects empowered by dense, distributed tactile sensing at the fingertips. To cope with complex object geometries, this study employs the point cloud classification method *PoinTacNet* for the classification of tactile information. The computational pipeline was devised in accordance with human behavior. A sim-to-real approach was employed to accommodate the data-efficient model training. Ten objects were classified with an accuracy of 83.89%. Furthermore, the implementation of an effective human haptic exploration strategy by re-grasping allows for noticeable improvements in classification accuracy; that is, the accumulation of class likelihood distribution increases the certainty of in-hand objects. Lastly, the point cloud registration algorithm, yielded pose estimation results with an average RMSE of 5.03° for angular errors and an average translational error of 2.41 mm.

Our method can be beneficial in various collaborative and manipulation tasks where the identification and pose of small objects are essential (e.g., assembly tasks involving small bolts). Adopting haptic exploration and class likelihood interpretation boosts decision reliability in performing tasks in real-world scenarios. Nevertheless, one limitation of the current work is the fixed object assumption. In situations where objects are not anchored, unexpected grasping configurations may occur, reducing the effectiveness of re-grasping. While this aspect is beyond the scope of our current work, it presents an interesting avenue for future research, potentially leading to greater robustness and adaptability in our proposed methodology.

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